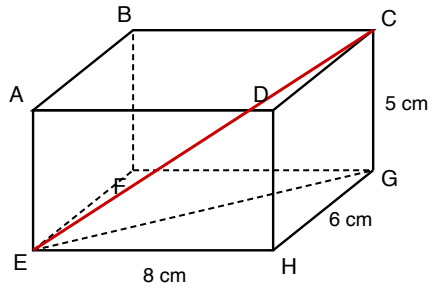
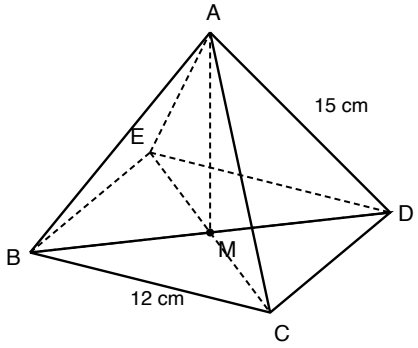
Shape and Space 9 — 3D trigonometry

Calculator allowed. Marks in brackets. ★ = harder question. Give lengths to 3 s.f. and angles to 1 d.p. unless told otherwise. Diagrams are not drawn accurately.

1 ABCDEFGH is a cuboid with EH = 8 cm, HG = 6 cm and CG = 5 cm. Find **a** the length EG **b** the angle between the diagonal EC and the base EFGH. [6]



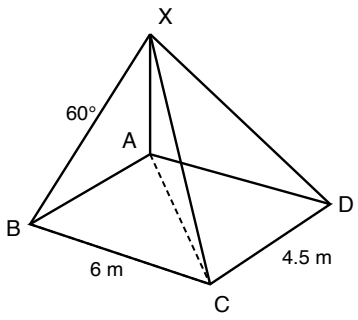
2 ABCDE is a square-based pyramid. The base BCDE is horizontal with side 12 cm, and AB = AC = AD = AE = 15 cm. M is the centre of the base, so AM is vertical. [13]



a Calculate **i** BD **ii** BM **iii** AM. **b** Calculate the angle that AD makes with the base.

c Calculate the angle between AM and the face ABC.

3 The diagram shows the roof of a kiosk. ABCD is a horizontal rectangle with AB $= 6$ m and BC $= 4.5$ m. The point X is vertically above A, and the edge XB makes an angle of 60° with the horizontal. Calculate the angle that the ridge XC makes with the horizontal. [6]



Total: 25 marks

Check: angle between a line and a plane: drop a perpendicular from the top of the line to the plane, join the foot to where the line meets the plane, use that right-angled triangle · redraw each 2D triangle with its lengths · the base diagonal of a cuboid/square needs Pythagoras first · keep full calculator values between steps.